
Practice 2 questions

For each of the following eight questions, select the most appropriate option by **filling the corresponding circle completely**.

Question 1: What is the primary goal of data science?

- ☐ Predicting future events with 100% accuracy
- ☐ Learning about the world from data using computation
- ☐ Creating complex algorithms for data manipulation
- ☐ Generating random datasets for experimentation

Question 2: Which of the following represent the stages of the Data Science Life Cycle?

- ☐ Data Mining, Data Validation, Model Deployment, Model Evaluation
- ☐ Problem Formulation, Data Acquisition, EDA, Inference and Prediction
- ☐ Data Visualization, Data Cleansing, Model Training, Model Testing
- ☐ Data Storage, Data Retrieval, Data Preprocessing, Model Interpretation

Question 3: Which components are typically involved in the design of a robotic agent?

- ☐ Sensors, Environment, Programming Languages
- ☐ Cameras, Motors, Software Libraries
- ☐ Sensors, Actuators, Environment
- ☐ Hardware, Software, Network Connectivity

Question 4: Which of the following accurately describes Python programming language?

- ☐ Compiled, specialized, low-level syntax with strict semantics
- ☐ Interpreted, domain-specific, high-level syntax with dynamic semantics
- ☐ Interpreted, general-purpose, high-level language with easy syntax and dynamic semantics
- ☐ Compiled, general-purpose, low-level syntax with strict semantics

Question 5: Which of the following statements accurately compare NumPy arrays with simple Python lists?

- ☐ NumPy arrays are slower and less versatile than Python lists.
- ☐ Python lists are more efficient in handling N-dimensional arrays compared to NumPy.
- ☐ NumPy arrays are the fundamental package for scientific computing, offering speed and versatility not found in Python lists.
- ☐ Python lists provide better vectorization, indexing, and broadcasting capabilities compared to NumPy.

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Question 6: Which of the following statements accurately differentiate between NumPy and Pandas libraries?

- NumPy is designed for working with tabular or heterogeneous data, while Pandas is best suited for homogenous numerical data.
- Pandas is derived from Panel Data and is primarily used for numerical computations, while NumPy is a data analysis library.
- Pandas adopts many coding idioms from NumPy, but the biggest difference lies in their suitability: Pandas for tabular or heterogeneous data, and NumPy for homogenous numerical data.
- NumPy offers fast and convenient data cleaning and analysis in Python, while Pandas is designed for working with specialized numerical arrays.

Question 7: What are the fundamental data structures in the Pandas library?

- Series, DataFrame, Vector
- Array, DataFrame, Indexer
- Series, Data Table, Index
- Series, DataFrame, Index

Question 8: Which statements accurately describe slicing methods in Pandas DataFrame?

- The `loc` method selects items by label, where the first argument represents rows and the second argument represents columns. The `iloc` method selects items by number, following the same order of arguments as `loc`. The square brackets `[]` can only take one argument, which may be a slice of row numbers, a list of column labels, or a single column label.
- The `iloc` method selects items by label, while the `loc` method selects items by number. The square brackets `[]` can take two arguments: a slice of row numbers and a list of column labels.
- The `loc` method selects items by number, while the `iloc` method selects items by label. The square brackets `[]` can only take one argument, which must be a list of column labels.
- The `loc` method selects items by label, where the first argument represents columns and the second argument represents rows. The `iloc` method selects items by number, following the same order of arguments as `loc`. The square brackets `[]` can only take one argument, which may be a slice of row numbers, a list of column labels, or a single column label.

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Question 9: How can I extract the even rows from the DataFrame named **babynames_first_10_rows** using a Boolean array as input for selection?

	State	Sex	Year	Name	Count
0	CA	F	1910	Mary	295
1	CA	F	1910	Helen	239
2	CA	F	1910	Dorothy	220
3	CA	F	1910	Margaret	163
4	CA	F	1910	Frances	134
5	CA	F	1910	Ruth	128
6	CA	F	1910	Evelyn	126
7	CA	F	1910	Alice	118
8	CA	F	1910	Virginia	101
9	CA	F	1910	Elizabeth	93

Even Rows

	State	Sex	Year	Name	Count
0	CA	F	1910	Mary	295
2	CA	F	1910	Dorothy	220
4	CA	F	1910	Frances	134
6	CA	F	1910	Evelyn	126
8	CA	F	1910	Virginia	101

Question 10: Which of the following pandas statements returns a DataFrame of the first 3 baby names with Count > 250.

- ☐ `babynames.iloc[[0, 233, 484], [3, 4]]`
- ☐ `babynames.loc[[0, 233, 484]]`
- ☐ `babynames.loc[babynames["Count"] > 250, ["Name", "Count"]].head(3)`
- ☐ `babynames.loc[babynames["Count"] > 250, ["Name", "Count"]].iloc[0:2, :]`

	State	Sex	Year	Name	Count
0	CA	F	1910	Mary	295
1	CA	F	1910	Helen	239
2	CA	F	1910	Dorothy	220
3	CA	F	1910	Margaret	163
4	CA	F	1910	Frances	134
5	CA	F	1910	Ruth	128
6	CA	F	1910	Evelyn	126

First 3 baby names with count > 250

	Name	Count
0	Mary	295
233	Mary	390
484	Mary	534

Question 11: Boolean array selection can be useful, but it often results in overly verbose code for complex conditions. What alternative does Pandas provide to simplify following process?

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```
babynames[(babynames["Name"] == "Bella") |  
            (babynames["Name"] == "Alex") |  
            (babynames["Name"] == "Ani") |  
            (babynames["Name"] == "Lisa")]
```

Question 12: How can we find the longest names in the pandas DataFrame named `babynames_ca`?

Required output.

	State	Sex	Year	Name	Count
313143	CA	M	1989	Franciscojavier	6
333732	CA	M	1997	Ryanchristopher	5
330421	CA	M	1996	Franciscojavier	8
323615	CA	M	1993	Johnchristopher	5
310235	CA	M	1988	Franciscojavier	10

Question 13: How can we find the best result by party in the pandas DataFrame named `elections` to achieve the required output as shown below?

	Year	Candidate	Popular vote	Result	%
Party					
American	1856	Millard Fillmore	873053	loss	21.554001
American Independent	1968	George Wallace	9901118	loss	13.571218
Anti-Masonic	1832	William Wirt	100715	loss	7.821583
Anti-Monopoly	1884	Benjamin Butler	134294	loss	1.335838
Citizens	1980	Barry Commoner	233052	loss	0.270182
Communist	1932	William Z. Foster	103307	loss	0.261069
Constitution	2008	Chuck Baldwin	199750	loss	0.152398
Constitutional Union	1860	John Bell	590901	loss	12.639283
Democratic	1964	Lyndon Johnson	43127041	win	61.344703

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Question 19: Given a logistic regression model fitted without an intercept using training data to predict the probability of a team winning a game based on GOAL_DIFF (first feature) and number of free throws (second feature), how would you predict the probability that a new team wins their game?

$$\hat{\theta}^T = [0.1 \quad -0.5] \quad x^T = [15 \quad 1]$$

Question 20: What problems might arise from using MSE loss with logistic regression?